# YANG Xuehuan

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## Career Summary

- **Key Skills**: Physical AI · Embodied AI · Autonomous Systems · SLAM · Sensor Fusion · VLA Models
- **Summary**: Physical AI Engineer with hands-on experience since 2017 across the NTU–Volvo–LTA autonomous bus, NTU–DSTA autonomous vehicle security, PSA port automation, and robotic systems at ST Engineering. I build SLAM, VLA and multi-robot systems, validated in simulation and deployed on real robots, to deliver safe and reliable autonomy.

## Working Experience

- **Singapore Technologies Engineering (ST Engineering IHQ)** — Singapore
  *Physical AI Engineer, Autonomous System of Systems, Group Engineering Center (GEC)* — *Apr 2026 - Present*
  - **Multi-Robot Framework**: Contributed to a software framework for multi-robot systems. Developed SLAM for robot navigation, built and integrated the execution and communication layers (ROS 2), and validated the system in simulation (Isaac Sim, AirSim) before real-world deployment.
  - **VLA Navigation**: Developed and tested VLA models for autonomous robot navigation. Set up OmniVLA pipelines, collected real-world robot data, fine-tuned the models, and validated them on physical robots.
- **SingPilot** — Singapore
  *AI Engineer, Autonomous Perception Team* — *Feb 2023 - Oct 2025*
  - **PSA EPM, Singapore's First Battery Charging and Swapping System (BCSS)**: Deployed and integrated the first XCMG Electric Prime Mover (EPM) at PSA. Worked with the BCSS team to integrate vehicle communication with the charging and battery-swapping processes. Explored LLM-driven log parsing to extract and interpret fault codes from raw CAN data.
  - **PSA HPM, Singapore's First Hydrogen Prime Mover**: Deployed and customized the first XCMG Hydrogen Prime Mover (HPM) at Singapore Port (Demo). Integrated vehicle charging protocols and the hydrogen refuelling process with NTU teams. Performed CAN bus data acquisition, fault code analysis, and on-site troubleshooting, and wrote technical reports with optimization recommendations.
  - **PSA APM**: Worked with the perception team to improve the accuracy and efficiency of autonomous driving detection models. Validated container and pedestrian detection modules on Terberg prime movers in real-world port operations. Maintained technical documentation aligned with system integration standards. Co-developed a pilot voice-assisted testing tool that combined ASR commands with LLM reasoning to streamline on-site validation.
  - **PSA-NTU Magnetic Localization**: Integrated autonomous driving software with terminal tractors and verified hardware–software compatibility for an ambient magnetic field positioning system. Collected and mapped magnetic field data across port operating areas, and tested sensor stability and localization performance in GNSS-degraded, structurally repetitive container yards.
  - **NTU COMS**: Collected and analyzed localization and perception data from autonomous electric vehicles in port environments. Built ROS-based integration and validation workflows on Ubuntu, using scripting for automation and troubleshooting. Explored natural-language query interfaces to allow engineers to interact with test data and logs using conversational prompts powered by LLMs.
- **School of Computer Science and Engineering@NTU** — Singapore
  *Research Associate, Cyber Security Team* — *Feb 2022 - Dec 2022*
  - **Alibaba-NTU AV Security**: Designed and ran experiments on autonomous vehicle security, focusing on realistic, stealthy physical adversarial triggers against lane detection systems. Processed LiDAR and camera data in Python for perception analysis and scenario simulation.
  - **Apollo LGSVL Simulation**: Ran Baidu Apollo with the LGSVL simulator to test autonomous driving software against pedestrian obstacles and corner cases, validating attack models before real-world trials. Prepared and processed test data to support AV system evaluation and improvement.
- **Centre for Autonomous Robotics, EEE@NTU** — Singapore
  *Research Intern, Autonomous Vehicle Localization Team* — *Jan 2019 - Dec 2019*
  - **NTU-DSTA ACRONIS**: Collaborated with DSTA on a Ministry of Defence (MINDEF) project on the safety and cybersecurity of driverless vehicles. Assessed cybersecurity risks across hardware, software, and system-level components of the autonomous driving stack, and supported attack experiments to evaluate system vulnerabilities.
  - **NTU CETRAN**: Conducted GPS signal amplification tests and physical attack experiments to evaluate sensor vulnerabilities. Processed and classified LiDAR and GPS data with Python, applying algorithmic compensations to enhance system robustness. Successfully tested autonomous vehicle localization on CETRAN testbed.

- **Energy Research Institute @ NTU (ERI@N)** Singapore
  *Research Intern, Future Mobility Team* *Sep 2017 - Dec 2018*
  - **NTU–JTC–Navya, Singapore's First Driverless 7m Bus**: Integrated ZED stereo camera-based lane detection into the vehicle's perception stack for real-time autonomous driving on campus, and validated it on-vehicle for reliable, safe navigation.
  - **NTU–Volvo–LTA, Singapore's First 12m Autonomous Electric Bus**: Developed targetless sensor calibration tools that need no checkerboard. Collected and processed 3D stereo camera data with CUDA for lane detection and obstacle recognition, and applied deep learning modules for real-time autonomous navigation.

## Education

- **Nanyang Technological University(NTU)** Singapore
  *Master of Engineering in Computer Science(SCSE)* *Jan 2020 - Dec 2021*
  - **Thesis**: Simulation-Based Perception Testing for Autonomous Vehicles(10356/154942)
  - **Research**: Assisted the team with the first physical attack on lane detection systems in autonomous driving, focusing on vulnerabilities and solutions.
- **Nanyang Technological University(NTU)** Singapore
  *Bachelor of Engineering in Electrical and Electronics (EEE)* *Aug 2016 - Dec 2019*
  - **FYP**: Interfacing and Testing of Localization Sensors on An Autonomous Vehicle(10356/136704)
  - **Specialisation**: Robotics and Control (Honours Distinction)

## Publications

- **ACMMM 2022 (CCF-A)**: Xingshuo Han, Guowen Xu, Yuan Zhou, **Xuehuan Yang**, Jiwei Li, Tianwei Zhang, Physical Backdoor Attacks to Lane Detection Systems in Autonomous Driving, Published in ACM International Conference on Multimedia (MM), October 2022 (arxiv 2203.00858)
- **TIFS 2022 (CCF-A)**: Jianfei Sun, Guowen Xu, Tianwei Zhang, **Xuehuan Yang**, Mamoun Alazab, Robert Deng, Verifiable, Fair and Privacy-preserving Broadcast Authorization for Flexible Data Sharing in Clouds, Published in IEEE Transactions on Information Forensics & Security(TIFS)(9969631)
- **TIFS 2023 (CCF-A)**: Jianfei Sun, Guowen Xu, Tianwei Zhang, **Xuehuan Yang**, Mamoun Alazab, Robert Deng, Privacy-Aware and Security-Enhanced Efficient Matchmaking Encryption, Published in IEEE Transactions on Information Forensics & Security(TIFS)(10180078)
- **TDSC 2023 (CCF-A)**: Jianfei Sun, Guowen Xu, Hongwei Li, Tianwei Zhang, Cong Wu, **Xuehuan Yang**, Robert Deng, Sanitizable Cross-domain Access Control with Policy-driven Dynamic Authorization, Published in IEEE Transactions on Dependable and Secure Computing(TDSC)(10891832)

## Award

- **Academic**: Graduated With Honours (Distinction), NTU
- **Competitions**: 3rd Prize, Chinese National High School Mathematics League; 3rd Prize, Chinese Chemistry Olympiad (Anhui Division); Merit Student, Hefei City
- **Driving Licence**: Singapore Class 3, 4, and 5
- **AV Operation**: Operated and tested autonomous cars, sweepers, and buses at NTU's CETRAN testbed; drove electric prime movers in port and on public roads.